Exercise list R9 — Mergers and acquisitions

Worked solutions

Statements are in `Listas/lista-r9-mergers-acquisitions.md`; each exercise is cited by number and printed page and is not restated. Short answers: `lista-r9-mergers-acquisitions-solutions-short.pdf`. Editions are the local copies: Higgins = Koski, Higgins & Mitton, *Analysis for Financial Management*, 13th ed. (2023); Grinblatt–Titman, *Financial Markets and Corporate Strategy*, 2nd ed. (2002); Amaro de Matos, *Theoretical Foundations of Corporate Finance* (2001); Tirole, *The Theory of Corporate Finance* (2005). Numbers that illustrate a conceptual answer are labelled "illustration" and are not part of the statement.

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1. True or false: valuation and acquisitions — Higgins 9.1, p. 378

What is asked. Five statements to classify as true or false, each with a short reason. (a) Whether acquisitions destroy value for the shareholders of the *acquired* (target) firm. (b) Which discount rate a DCF valuation of a target should use. (c) Whether a buyer should pay a larger control premium

for a well-run firm. (d) Whether liquidation value is a floor for the share price. (e) Whether a share price that management finds too low pushes management toward a buyout.

Approach. Higgins' framework for the market for control: the most a buyer can pay is the target's value under the buyer's control, V^{new}; the stand-alone (minority) value is V^{sq}; the control premium pays for $V^{\text{new}} - V^{\text{sq}}$, the value of changing how the firm is run. Discount rates follow the risk of the cash flows (value additivity), and every "floor" on a price requires someone with the power to enforce it.

Notation.

- V^{sq}: value of the target's equity under current management (the pre-bid market value of a minority stake).
- V^{new}: value under the acquirer's management; $V^{\text{new}} - V^{\text{sq}}$: value of control.
- r_T, r_A: costs of capital appropriate to the target's and the acquirer's cash flows.
- L: liquidation value per share; p: market price per share.

Step 1 — (a) Target shareholders. We look at who receives the premium. Bids are made well above the pre-bid price, and the target's price jumps on announcement (Figure 5.3 of Exercise 3: about +20 percentage points on the announcement days, $\approx 28\%$ cumulative). Target shareholders sell at V^{sq} + premium.

Answer (a). **False**: targets' shareholders gain on average (returns of 20% or more around the announcement); it is the *acquirers'* gains that are doubtful.

Step 2 — (b) Discount rate. We apply value additivity. The value of a stream of cash flows does not depend on who owns it, so the target's free cash flows FCF_t^T are discounted at the rate that reflects their own risk:

$$V^T = \sum_{t \geq 1} \frac{\mathbb{E}[FCF_t^T]}{(1 + r_T)^t}. \tag{1}$$

Using r_A would value a risky target too high when the acquirer is safer ($r_A < r_T$) and too low when it is riskier.

Answer (b). **False**: discount at the *target's* cost of capital (the risk of the cash flows being valued), not the acquirer's.

Step 3 — (c) Control premium. We compute the maximum premium as the value of control. Illustration: two targets each worth $V^{\text{sq}} = 100$; under the buyer the poorly run one is worth 140, the well run one 105 (little left to improve). The maximum premium in percent is

$$\frac{V^{\text{new}} - V^{\text{sq}}}{V^{\text{sq}}} = \frac{(140) - (100)}{100} = 40\% \quad \text{(poorly run)},$$
$$\frac{V^{\text{new}} - V^{\text{sq}}}{V^{\text{sq}}} = \frac{(105) - (100)}{100} = 5\% \quad \text{(well run)}.$$

Answer (c). **False**: the premium pays for improvements; a well-managed firm offers little to improve, so the premium should be *smaller*.

Step 4 — (d) Liquidation value. We ask who can force liquidation. A minority shareholder cannot sell the assets; only the controlling party (or, with dispersed ownership, incumbent management) can. If they prefer to continue (optimism, private benefits), the price can stay below L: illustration, $L = 50$ and $p = 40$, a gap of $50 - 40 = 10$ per share that only a buyer of control can capture.

Answer (d). **False**: liquidation value is a floor only if someone with control is willing to liquidate; otherwise the shares can trade below it (which is precisely what invites a raider).

Step 5 — (e) MBO motive. Management knows more than the market. Illustration from the book's answer: price \$30, management's estimate \$80, so buying the firm yields $80 - 30 = 50$ per share to the insiders.

Answer (e). **True**: an unusually low price relative to management's private valuation makes a management buyout attractive, and also creates a conflict of interest (managers gain from a low price).

Intuition. All five items read off one identity: price paid $= V^{\text{sq}}$+premium, and the premium is bounded by the value of control $V^{\text{new}} - V^{\text{sq}}$. Who captures that value depends on competition among bidders and on information; who can enforce a floor depends on control rights.

Common mistakes.

- Confusing target and acquirer returns: targets gain, acquirers roughly break even on average.
- Using the acquirer's WACC for the target because "the acquirer is paying"; the discount rate belongs to the cash flows.
- Treating liquidation value as an arbitrage floor; it is not, because liquidation requires control.

Book answer. True/false classification matches the book's answer (pp. 425–426): F, F, F, F, T.

Ref: Higgins Ch. 9, "The Market for Control" and "The Premium for Control", pp. 354–357; "The Empirical Evidence", p. 364; Figure 5.3, p. 169.

2. LBO candidates and management buyouts — Grinblatt & Titman 20.5 and 20.8, p. 731

What is asked. (20.5) Why LBOs cluster in industries with stable cash flows and non-specialized products. (20.8) Why an MBO improves incentives even though the same managers stay in charge.

Approach. An LBO replaces most equity by debt. Debt capacity is limited by expected distress costs, which depend on cash-flow volatility (risk of default) and on asset specificity (how much value is lost in liquidation or fire sales). For 20.8 the lever is ownership and the free-cash-flow discipline of debt (Jensen 1986): the agency problem is about the managers' *stake* and *discretion*, not their identity.

Notation.

- X: next-period operating cash flow; D: face value of debt; riskless debt requires $D \leq \min X$.
- s: managers' fraction of the equity; a \$1 increase in equity value gives managers $s \cdot 1$.

Step 1 — (20.5) Stability and debt capacity. We compare how much riskless debt two firms with the same mean cash flow can carry. Illustration: stable firm $X \in \{90, 110\}$, volatile firm $X \in \{40, 160\}$, both with mean 100. The largest face value repaid in every state is the worst cash flow:

$$D_{\text{stable}} = 100 - 10 = 90,$$
$$D_{\text{volatile}} = 100 - 60 = 40.$$

The stable firm can borrow more than twice as much without default risk, so the tax shield and the discipline of debt come with few expected distress costs.

Step 2 — (20.5) Specialization. Specialized assets and products lose value when the firm is distressed: customers fear for after-sales service and spare parts, suppliers ask for cash, and specialized assets fetch low prices because few buyers can use them (the Shleifer–Vishny point of Exercise 14). Expected distress costs are therefore larger for the same default probability.

Answer (20.5). Highly levered structures are cheap only when default is unlikely (stable cash flows) and costly default is mild (generic products and redeployable assets); both reduce the expected costs of financial distress that limit leverage, while the tax and free-cash-flow benefits remain.

Step 3 — (20.8) Same managers, new incentives. Illustration: before the buyout managers own $s = 5\%$; after it, $s = 15\%$ (Kaplan 1989: management ownership roughly triples). For each \$1 of value they create,

$$\text{before: } s \cdot 1 = 0.05, \qquad \text{after: } s \cdot 1 = 0.15, \qquad \frac{0.15}{0.05} = 3.$$

In addition, the heavy debt service pre-commits free cash flow, which removes the option to spend it on negative-NPV projects, and the buyout sponsor sits on a small active board.

Answer (20.8). The incentive problem is not that the managers are bad but that they own little and have discretion over free cash flow. An MBO raises their equity stake and commits cash to debt service and active owners, so the *same* managers face sharper incentives. Keeping them also keeps their firm-specific knowledge.

Intuition. An LBO trades financial flexibility for discipline. That trade pays where flexibility is least needed (steady, mature businesses) and where the discipline is most valuable (large free cash flow).

Common mistakes.

- Saying LBOs need "high growth"; growth firms need flexibility and have intangible, specific assets.
- Thinking an MBO's benefit requires replacing managers; it comes from ownership and debt, not new people.

Ref: Grinblatt & Titman Ch. 20 (LBOs and MBOs, financial synergies and disciplinary takeovers); Amaro de Matos §7.2, pp. 187–189.

3. Target stock prices around acquisition announcements — Higgins 5.10, p. 189

What is asked. (a) Whether a three-day price response is consistent with efficiency. (b) A trading rule that would profit from it. (c) What happens to the response if many traders use the rule. (d) How this shows inefficiencies "destroying themselves". (e) Whether a target trading above the bid proves inefficiency, and what explains it.

Approach. Semistrong efficiency: prices reflect all public information at once, so no rule based on public news earns abnormal returns. A predictable post-announcement drift is a public, exploitable pattern. The price above the bid is valued as an expectation over possible outcomes (counter-bids), which is the competition-among-bidders logic of the M&A models (Exercise 9).

Notation.

- CAR_t: cumulative percentage change in the target's price up to day t (Figure 5.3 values in the list).
- Index $100\,(1 + CAR_t)$: price relative to day -40.

Step 1 — (a) Efficiency. From Figure 5.3 the announcement move is from $\approx 8.5\%$ (day -1) to $\approx 28.5\%$ (day $+1$), a jump of $28.5 - 8.5 = 20$ points that happens almost instantly. If the same jump were spread over three days, the day-1 and day-2 returns would be predictable from public news already released.

Answer (a). No: a predictable response to public news violates semistrong-form efficiency.

Step 2 — (b) Trading rule. We compute the return of buying at the close of day 0. Suppose one third of the jump occurs on day 0. The index at the close of day 0 is

$$100 + 8.5 + \frac{20}{3} = 115.17, \tag{2}$$

and selling at the close of day 2 at $100 + 28.5 = 128.5$ yields

$$\frac{128.5}{115.17} - 1 = 11.6\% \tag{3}$$

in two days with no risk specific to the strategy beyond deal risk.

Answer (b). Buy target shares as soon as an offer is announced and sell when the price has finished adjusting (two days later).

Step 3 — (c)–(d) Self-destruction. Purchases at the close of day 0 push the day-0 price up; competition among traders moves the trade ever earlier until the whole adjustment happens on the announcement, i.e. the pattern of Figure 5.3.

Answer (c). The response would become (almost) instantaneous; the profit disappears.

Answer (d). The profit opportunity attracts the trading that eliminates it; this is why markets with many informed traders look efficient.

Step 4 — (e) Price above the bid. We value a target share as an expectation over outcomes. Illustration: bid \$60; with probability 0.3 a rival bids \$75, with 0.6 the bid succeeds at \$60, with 0.1 the deal fails and the price falls back to \$35:

$$\mathbb{E}[\text{value}] = (0.3)(75) + (0.6)(60) + (0.1)(35) = 22.5 + 36 + 3.5 = 62 > 60. \tag{4}$$

Answer (e). No: the market expects a higher (competing or revised) bid; the price above the offer is the expected value of the share, not a mistake.

Intuition. Event studies measure how fast public news enters prices. The jump on day 0 is the premium being capitalized, and a price above the bid is the market's forecast of a bidding contest.

Common mistakes.

- Reading the pre-announcement drift (days -40 to -1) as proof of inefficiency; it is consistent with leakage and rumours (and with insider trading, which semistrong efficiency does not rule out).
- Concluding in (e) that "price > bid" is irrational; it is a probability-weighted value.

Ref: Higgins Ch. 5, "How Rapidly Do Stock Prices Adjust to New Information?" and Figure 5.3, pp. 168–169.

4. Sources of LBO premiums — Amaro de Matos 7.2.4 #3, p. 189

What is asked. Explain where the large and lasting premiums received by pre-buyout public shareholders come from.

Approach. A premium is a transfer of part of the value $V^{\text{post}} - V^{\text{pre}}$ to the selling shareholders. List the sources of that difference: real gains (taxes, incentives, efficiency) and transfers from other claimants (bondholders, employees) or from information.

Notation.

- D: new permanent debt; τ: corporate tax rate; τD: present value of a perpetual tax shield (MM with taxes).
- V^{pre}, V^{post}: firm value before and after the buyout.

Step 1 — Tax benefits. We value the interest tax shield. Illustration: permanent new debt $D = 600$ and $\tau = 34\%$:

$$PV(\text{tax shield}) = \tau D = (0.34)(600) = 204. \tag{5}$$

Kaplan finds that pre-buyout shareholders' returns are positively related to these tax benefits.

Step 2 — Agency costs. Management ownership roughly triples (Kaplan 1989a), and debt service commits free cash flow that managers could otherwise waste (Jensen's free-cash-flow hypothesis); operating performance improves after the buyout.

Step 3 — Transfers. Bondholders and preferred stockholders lose at the announcement (the new debt is senior or riskier), and employment grows more slowly; these losses are small relative to the premiums.

Step 4 — Information and efficiency. Managers know more than outside shareholders, so a buyout can signal undervaluation; and private ownership speeds decisions and avoids disclosure to rivals.

Answer (#3). Tax savings from high leverage; lower agency costs (managerial ownership and the discipline of debt on free cash flow); wealth transfers from bondholders, preferred stockholders and employees; managers' private information (undervaluation); efficiency gains of private ownership.

Intuition. The premium is sustainable when it rests on real value creation (taxes, incentives, efficiency); transfers and information effects explain part of it but are too small on their own.

Common mistakes.

- Listing only taxes: Kaplan shows post-buyout returns follow operating improvements, not taxes.
- Treating bondholder losses as the main source; the evidence finds them small.

Book answer. Matches the book's answer (p. 271): taxes, agency costs, transfers from bondholders, preferred stockholders and employees, information, efficiency.

Ref: Amaro de Matos §7.2.3, pp. 187–189.

5. Grossman–Hart with a toehold — Amaro de Matos 8.1.5 #1, p. 205

What is asked. Show that a bidder who already owns shares of the target can profit from a bid at a tender price $p \geq v$, although without shares such a bid makes a loss.

Approach. This is the Grossman–Hart *free-rider problem*: atomistic shareholders tender only if $p \geq v$, because a non-tendering shareholder keeps a share worth v after a successful bid. With no toehold the raider then buys the whole improvement at its full value and loses the cost c. A toehold earns the improvement on the shares already owned, which free riding cannot take away (Tirole §11.5.2.2).

Notation.

- $q = f(a^\circ)$: value per share under incumbent management (status quo); $v = \max_a f(a) + Z$: value under the raider, $Z > 0$; shares normalized to 1.
- p: tender price per share; $c \geq 0$: cost of the raid; $\theta \in (0, \frac{1}{2})$: toehold (the book's answer calls it x).

Step 1 — Why shareholders need $p \geq v$. A shareholder's tender decision does not change the outcome (Assumption 8.1.1). If the bid succeeds, keeping a share yields v and tendering yields p; so tendering is optimal only if

$$p \geq v. \tag{6}$$

Step 2 — No toehold. The raider buys all shares at p and owns a firm worth v:

$$\Pi_r(\theta = 0) = v - p - c \leq v - v - c = -c < 0 \quad \text{when } p \geq v. \tag{7}$$

Step 3 — With a toehold. The raider already owns θ and buys the remaining $1-\theta$ at p. After the bid he owns the whole firm, worth v. The book counts his end-of-raid wealth minus the cash spent:

$$\Pi_r = v - p(1-\theta) - c. \tag{8}$$

Setting (8) positive and dividing by $p > 0$,

$$\begin{aligned} v - p + p\theta - c &> 0 \\ p\theta &> p - v + c \\ \theta &> 1 + \frac{c-v}{p}. \end{aligned} \tag{9}$$

At $p = v$ the threshold is $1 + (c-v)/v = c/v$, which is below 1 whenever $c < v$: some toehold suffices.

Step 4 — Incremental profit. Before bidding, the toehold is worth θq; the raid's *net* gain subtracts it:

$$\begin{aligned} \Delta\Pi_r &= v - p(1-\theta) - c - \theta q \\ &= \theta(v-q) - (1-\theta)(p-v) - c. \end{aligned} \tag{10}$$

At $p = v$ this is $\theta(v-q) - c$, positive if $\theta > c/(v-q)$. Illustration: $v = 100$, $q = 80$, $c = 1$, $\theta = 0.1$, $p = 100$:

$$\begin{aligned} \Pi_r &= (100) - (100)(1-0.1) - (1) = 100 - 90 - 1 = 9, \\ \Delta\Pi_r &= (0.1)(100-80) - (1) = 2 - 1 = 1 > 0, \end{aligned}$$

while without a toehold $\Pi_r = 100 - 100 - 1 = -1$. Solving $\Delta\Pi_r = 0$ in (10) for p, the highest profitable bid is $p = v + [\theta(v-q) - c]/(1-\theta) = 100 + 1/0.9 = 101.11$.

Answer (#1). With a toehold θ, $\Pi_r = v - p(1-\theta) - c > 0$ iff $\theta > 1 + (c-v)/p$ (at $p = v$: $\theta > c/v$); net of the toehold's initial value the bid pays iff $\theta(v-q) > c + (1-\theta)(p-v)$. The raider profits from the value created on the shares he already holds.

Intuition. Free riding lets the small shareholders capture all the value created on *their* shares; it cannot touch the value created on the raider's own shares. A toehold therefore restores an incentive to raid even at a price that leaves the tendering shareholders indifferent.

Common mistakes.

- Forgetting that tendering shareholders compare p with v (post-raid value), not with q.
- Counting the raider's profit on the toehold as θv instead of $\theta(v-q)$: the toehold was already worth θq.

Book answer. Matches the book's answer (p. 271): profit $v - p(1-x) - c > 0 \iff x > 1 + (c-v)/p$. The book does not subtract the toehold's pre-bid value; Step 4 adds that refinement.

Ref: Amaro de Matos §8.1.1, pp. 193–194; Tirole §11.5.2.2, p. 433.

6. News Corp. and Dow Jones — Higgins 9.3, p. 379

What is asked. (a) The cost of the acquisition to News Corp.'s shareholders. (b) The value News Corp. placed on control of Dow Jones.

Approach. Buying all the shares of a company means buying its assets and *assuming* its liabilities, so the cost of the acquisition is the price paid for equity plus the (market value, approximated by book value) of the interest-bearing debt taken over. The value of control is the premium over the pre-bid (minority) market value.

Notation.

- $p_b = \$60$: bid per share; $p_0 = \$33$: pre-bid price; $n = 82$ million shares.
- $D = \$1.46$ billion: interest-bearing liabilities (book value, used as proxy for market value).

Step 1 — (a) Equity paid. We multiply the bid by the shares outstanding:

$$p_b n = (60)(82 \text{ million}) = \$4.92 \text{ billion}. \tag{11}$$

Step 2 — (a) Add the assumed debt. The buyer becomes responsible for Dow Jones's debt:

$$\text{Cost} = p_b n + D = (4.92) + (1.46) = \$6.38 \text{ billion}. \tag{12}$$

Answer (a). About \$6.38 billion (an estimate because book debt approximates market debt).

Step 3 — (b) Value of control. We compute the premium over the pre-bid price, per share and in total:

$$\begin{aligned} (p_b - p_0)\, n &= (60 - 33)(82 \text{ million}) = (27)(82 \text{ million}) = \$2.214 \text{ billion}, \\ \frac{p_b - p_0}{p_0} &= \frac{27}{33} = 81.8\%. \end{aligned} \tag{13}$$

Answer (b). About \$2.21 billion, an 81.8% premium over the pre-bid equity value of $33 \times 82 =$ \$2.706 billion.

Intuition. The pre-bid price is the value of a minority stake under current management; News Corp. paid \$2.21 billion more because it expected to run Dow Jones differently (or to enjoy private benefits).

Common mistakes.

- Reporting only the \$4.92 billion paid for equity: the assumed debt is part of the price.
- Computing the control value from total cost (\$6.38 billion) instead of the premium over the pre-bid equity value.

Book answer. Matches the book's answer (p. 426): \$6.38 billion and \$2.21 billion.

Ref: Higgins Ch. 9, "The Premium for Control", pp. 354–357.

7. Procureps: EPS dilution as a merger criterion — Higgins 9.9, p. 381

What is asked. (a) Fill in the table: market values, prices, the maximum number of shares Procureps (P) can issue without diluting its EPS of \$2, their value and the implied maximum premium for V1 and for V2. (b) What this says about the "avoid EPS dilution" rule.

Approach. In an exchange-of-shares merger with no synergies, the combined earnings are the sum of the earnings. Keeping EPS at \$2 caps the combined share count; the extra shares are worth P's price; the premium compares that value with the target's market value. The rule depends only on P/E ratios, not on value creation.

Notation.

- $E_P = 2$, $E_{V1} = E_{V2} = 1$: earnings (\$ million); P/E: 30, 8, 35; one million shares each.
- $n^{\max}$: maximum combined shares with EPS ≥ 2; $\Delta n = n^{\max} - 1$: new shares; p_P: P's share price.

Step 1 — Market values and prices. Market value = P/E × earnings, and price = value / shares:

$$\begin{aligned} MV_P &= (30)(2) = 60, & p_P &= 60/1 = \$60, \\ MV_{V1} &= (8)(1) = 8, & p_{V1} &= \$8, \\ MV_{V2} &= (35)(1) = 35, & p_{V2} &= \$35. \end{aligned}$$

Step 2 — Maximum shares. Combined earnings are $2 + 1 = 3$ in either merger. EPS stays at 2 if

$$\frac{3}{n^{\max}} = 2 \;\Rightarrow\; n^{\max} = \frac{3}{2} = 1.5 \text{ million}, \qquad \Delta n = 1.5 - 1 = 0.5 \text{ million}. \tag{14}$$

Step 3 — Value of the new shares and premium. At P's price the 0.5 million new shares are worth $(0.5)(60) = \$30$ million. The maximum premium over each target's market value is

$$\text{V1: } \frac{30}{8} - 1 = 3.75 - 1 = 275\%, \tag{15}$$

$$\text{V2: } \frac{30}{35} - 1 = 0.857 - 1 = -14.3\%. \tag{16}$$

Company	P	V1	P+V1	V2	P+V2
Earnings ($m)	2	1	3	1	3
P/E	30	8		35	
Market value ($m)	60	8		35	
Shares (m)	1	1	1.5	1	1.5
EPS ($)	2	1	2	1	2
Price ($)	60	8		35	
Max. new shares (m)			0.5		0.5
Value of new shares ($m)			30		30
Max. premium (%)			275		−14

Answer (a). Procureps can pay up to a 275% premium for V1 but must buy V2 at a 14% *discount* to avoid EPS dilution.

Step 4 — (b) What the rule does to value. With no synergies, value is conserved. Buying V2 at its market price issues $35/60 = 0.583$ million shares; EPS falls to $3/1.583 = \$1.89$, but the share price is unchanged:

$$\frac{60 + 35}{1 + 0.583} = \frac{95}{1.583} = \$60.$$

Buying V1 at the "allowed" 275% premium keeps EPS at \$2 but transfers $30 - 8 = \$22$ million to V1's holders: P's price falls to $(60 + 8)/1.5 = \$45.33$, a loss of \$14.67 per share.

Answer (b). The criterion is shortsighted: it favours low-P/E, low-growth targets and penalizes high-growth ones, and it can approve value-destroying deals while rejecting value-neutral ones. Value, not EPS, is the test.

Intuition. EPS accretion is an artefact of exchanging high-P/E paper for low-P/E earnings; the market prices the combined firm at a blended P/E, so EPS moves while value does not.

Common mistakes.

- Valuing the new shares at the target's price instead of the acquirer's.
- Reading EPS dilution as value destruction (the V2 case shows EPS down, price unchanged).

Book answer. Matches the book's answer (p. 427): 275% and −14%.

Ref: Higgins Ch. 9, "Avoiding Dilution in Earnings per Share", p. 362.

8. Hansen: cash versus stock with a uniform target value — Amaro de Matos 8.2.6 #3, p. 223

What is asked. Show that with $v \sim U(0,100)$ and $w(v) = 1.5v$ no cash offer benefits both sides, while an offer of a fraction β of the merged firm's shares is accepted by every target type and still benefits the acquirer whenever its own value x is below 150.

Approach. Hansen's model is a lemons problem: the target knows v, the acquirer does not. A cash offer C is accepted only by targets with $v \le C$ (adverse selection), so the acquirer's gain is computed *conditional on acceptance*. A stock offer makes the target's payment contingent on what the target is worth (the target receives a share of $x + w(v)$), which weakens the adverse selection: the means of payment matters under asymmetric information.

Notation.

- $v \in (0,100)$: target's stand-alone value, privately known to the target; $F(v) = v/100$, $f(v) = 1/100$.
- $w(v) = 1.5v$: target's value to the acquirer; $x > 0$: acquirer's value, publicly known.
- C: cash offer; $\beta \in (0,1)$: fraction of the merged firm offered; v^*: highest accepting type.
- $E(w \mid C) = \mathbb{E}[w(v) \mid v \le C]$: value to the acquirer of the targets that accept.

Step 1 — Who accepts a cash offer. The target sells for C only if $C \ge v$, so the accepted types are $v \in (0, C]$ with probability $F(C) = C/100$ (for $C \le 100$).

Step 2 — Value of what the acquirer gets. Conditional on $v \le C$, v is uniform on $(0, C)$ with mean $C/2$, so

$$E(w \mid C) = 1.5\,\mathbb{E}[v \mid v \le C] = 1.5 \cdot \frac{C}{2} = 0.75\,C. \tag{17}$$

Step 3 — Cash gain. Substituting (17) into $E(\Pi_r \mid C) = F(C)[E(w \mid C) - C]$,

$$\begin{aligned} E(\Pi_r \mid C) &= \frac{C}{100}\left[(0.75C) - C\right] \\ &= -\frac{0.25\,C^2}{100} < 0 \quad \text{for every } C \in (0,100]. \end{aligned} \tag{18}$$

(At $C = 100$, where every type accepts, the loss is -25.) The acquirer always pays C for assets worth $0.75C$ to it: although $w(v) = 1.5v > v$ creates gains from trade for every type, no cash offer is profitable.

Answer (cash). No mutually beneficial cash trade exists: $E(\Pi_r \mid C) = -0.0025\,C^2 < 0$ for all C.

Step 4 — Who accepts a stock offer. A target of type v holding β of the merged firm gets $\beta[x + w(v)]$; it accepts iff

$$\beta(x + 1.5v) \ge v \iff \beta \ge \frac{v}{x + 1.5v}. \tag{19}$$

The right-hand side increases in v: its derivative is

$$\frac{d}{dv}\frac{v}{x+1.5v} = \frac{(x+1.5v) - 1.5v}{(x+1.5v)^2} = \frac{x}{(x+1.5v)^2} > 0,$$

which is Hansen's condition $dw/dv = 1.5 < (x + 1.5v)/v$. So the hardest type to please is $v = 100$, and every type accepts iff

$$\beta(x + 150) \ge 100 \iff \beta \ge \frac{100}{x + 150}. \tag{20}$$

Step 5 — Acquirer's participation. If every type accepts, the acquirer gets $w(v)$ with mean $\mathbb{E}[1.5v] = 1.5 \times 50 = 75$, and keeps $1-\beta$ of the merged firm. Condition (19) of the model, (8.12), is

$$(1-\beta)(x+75) \geq x \iff \beta \leq 1 - \frac{x}{x+75} = \frac{75}{x+75}. \tag{21}$$

Step 6 — Both conditions. A β exists iff the lower bound (20) does not exceed the upper bound (21). Cross-multiplying (both denominators positive):

$$\begin{aligned} \frac{100}{x+150} &\leq \frac{75}{x+75} \\ 100(x+75) &\leq 75(x+150) \\ 100x + 7500 &\leq 75x + 11250 \\ 25x &\leq 3750 \\ x &\leq 150. \end{aligned} \tag{22}$$

With $x < 150$ the inequality is strict, so some β gives the acquirer a strict gain. Example $x = 100$: $\beta \in [100/250,\ 75/175] = [0.4,\ 0.4286]$; at $\beta = 0.4$ the acquirer's gain is $(1-0.4)(100+75) - 100 = 105 - 100 = 5$.

Step 7 — Necessity (partial-acceptance stock offers). An offer accepted only by $v \leq v^*$ sets $\beta = v^*/(x+1.5v^*)$ and yields conditional value $E(w \mid v^*) = 0.75v^*$, so the gain is $F(v^*)\{(1-\beta)(x+0.75v^*) - x\}$. Since $1-\beta = (x+0.5v^*)/(x+1.5v^*)$,

$$\begin{aligned} (1-\beta)(x+0.75v^*) - x &= \frac{(x+0.5v^*)(x+0.75v^*) - x(x+1.5v^*)}{x+1.5v^*} \\ &= \frac{x^2 + 1.25xv^* + 0.375v^{*2} - x^2 - 1.5xv^*}{x+1.5v^*} \\ &= \frac{v^*(0.375v^* - 0.25x)}{x+1.5v^*}, \end{aligned}$$

positive iff $v^* > \frac{2}{3}x$. As $v^* \leq 100$, some stock offer gains iff $x < 150$: the bound is also necessary.

Answer (stock). Any $\beta \in [100/(x+150),\ 75/(x+75)]$ is accepted by all types and benefits the acquirer; the interval is non-empty iff $x \leq 150$ (strict gain for $x < 150$), and no stock offer gains for $x \geq 150$.

Intuition. Cash is a fixed payment: only targets worth less than the offer accept, so the acquirer buys lemons. Stock pays each target in proportion to the merged value, which rises with $w(v)$; part of the payment adjusts with the target's quality. The adjustment is diluted when the acquirer is large, because then $\beta(x+w)$ moves little with v; hence the upper bound on x.

Common mistakes.

- Evaluating the cash offer at the unconditional mean $\mathbb{E}[w] = 75$ instead of the mean among targets that accept ($0.75C$): the winner's curse.
- Checking the target's condition at the mean v instead of at the highest accepting type.
- Reading the book's "$p \leq 75/(x+75)$" literally; the p is a misprint for β.

Book answer. Matches the book's answer (p. 276): the acquirer gets $0.75C$ for C; stock works when $\beta(x+150) \geq 100$ and $(1-\beta)(x+75) \geq x$, consistent iff $x \leq 150$.

Ref: Amaro de Matos §8.2.2, pp. 207–210; Hansen (1987).

9. Fishman (1988): why the preemptive signal is not fully revealing — Amaro de Matos 8.2.6 #1, p. 223

What is asked. Why bidder 1's opening bid in Fishman's equilibrium reveals only a range of v_1, and how the equilibrium would change if bidder 2 could observe v_1.

Approach. Preemptive bidding is signalling: a high opening bid deters bidder 2 from paying c_2 to enter an auction. Deterrence only requires bidder 2 to believe that v_1 is high enough on average (above the threshold r); once that belief is induced, a higher type gains nothing by bidding more. Compare with the complete-information benchmark where no signal is needed.

Notation.

- v: target's stand-alone value; v_i: bidder i's valuation, density f_i on $[l, h]$; c_2: bidder 2's cost of learning v_2.
- $g(v_1) \equiv \mathbb{E}_{v_2}[\Pi_2(v_1, \tilde{v}_2, \cdot, 1)] + c_2 = \mathbb{E}[\max(\tilde{v}_2 - v_1, 0)]$: bidder 2's gross expected gain from an auction against v_1, decreasing in v_1.
- r: preemption threshold, $\mathbb{E}[g(\tilde{v}_1) \mid \tilde{v}_1 \geq r] = c_2$; w: complete-information entry cutoff, $g(w) = c_2$.
- $\bar{p}(v_1) = \mathbb{E}_{v_2} \min[\max(v, \tilde{v}_2), v_1]$: the highest preemptive bid type v_1 is willing to make.

Step 1 — Bidder 2's payoff. From (8.10) with $d = 1$, $\Pi_2 = v_2 - \min(v_2, v_1) - c_2 = \max(v_2 - v_1, 0) - c_2$: in the ascending auction bidder 2 wins when $v_2 > v_1$ and pays v_1. Hence its expected payoff is $g(v_1) - c_2$, and g falls as v_1 rises.

Step 2 — What the equilibrium reveals. Bidder 1 offers $\bar{p}(r)$ if $v_1 \geq r$ and v if $v_1 < r$. Bidder 2 sees one of two messages: "$v_1 \in [v, r)$" or "$v_1 \geq r$". All types above r make the *same* bid. They do not separate because bidder 2's decision depends only on its belief about v_1, and the pooled belief $\mathbb{E}[g \mid v_1 \geq r] = c_2$ already makes it stay out; bidding more costs a type above r money and changes nothing.

Answer (why). The equilibrium is a two-message (partition) equilibrium: bidder 2 learns only whether $v_1 < r$ or $v_1 \geq r$, not v_1 itself, so it is not fully revealing.

Step 3 — Complete information. If bidder 2 observed v_1, it would enter iff $g(v_1) \geq c_2$, i.e. iff $v_1 < w$ with $g(w) = c_2$. Bidder 1 would no longer need a costly signal; he would open at v (the lowest acceptable price) and let bidder 2 decide on the true v_1. Because g is decreasing and $\mathbb{E}[g \mid v_1 \geq r] = c_2$, the value at the bottom of the pool is above the pool average: $g(r) > c_2 = g(w)$, hence $w > r$.

Step 4 — Illustration. Take $v_1, v_2 \sim U[0, 1]$, $v = 0.6$ (so $\mathbb{E}[\tilde{v}_i] = 0.5 < v$) and $c_2 = 0.02$. Then $g(v_1) = \int_{v_1}^{1} (t - v_1)\, dt = (1 - v_1)^2/2$.

$$\text{cutoff } w: \quad \frac{(1-w)^2}{2} = 0.02 \;\Rightarrow\; 1 - w = \sqrt{0.04} = 0.2 \;\Rightarrow\; w = 0.8,$$

$$\text{threshold } r: \quad \frac{1}{1-r}\int_r^1 \frac{(1-t)^2}{2}\, dt = \frac{(1-r)^2}{6} = 0.02 \;\Rightarrow\; 1 - r = \sqrt{0.12} = 0.3464 \;\Rightarrow\; r = 0.6536.$$

At $v_1 = r$, $g(r) = 0.12/2 = 0.06 > 0.02$. The preemptive bid is

$$\begin{aligned}\bar{p}(r) &= v \cdot \Pr(\tilde{v}_2 < v) + \int_v^r t\, dt + r \cdot \Pr(\tilde{v}_2 > r) \\ &= (0.6)(0.6) + \frac{r^2 - 0.36}{2} + r(1 - r) = 0.36 + 0.0336 + 0.2264 = 0.62.\end{aligned}$$

Types $v_1 \in [0.6536,\ 0.8)$ deter bidder 2 by pooling at 0.62 but would face entry under observability.

Answer (observable v_1). Bidder 1 bids v; bidder 2 competes iff $v_1 < w$, with $w > r$. Types in $[r, w)$ are deterred in the signalling equilibrium but not under observability: preemption is more frequent with private information, and the target receives a premium ($\bar{p}(r) > v$) only because of it.

Intuition. Signalling is needed only to the extent that it changes the rival's action, and that action is binary (enter or not). A binary decision can be steered by a binary message, so the equilibrium pools types. Private information lets weak-but-above-r types free ride on strong types' deterrence.

Common mistakes.

- Expecting separation because types differ: separation requires each type's payoff to depend on the rival's *exact* belief, which it does not here.
- Computing r from $g(r) = c_2$ (the marginal type) instead of the average over $v_1 \geq r$.

Book answer. Matches the book's answer (p. 276): bidder 2 learns only $v \leq v_1 < r$ or $v_1 \geq r$; with observable v_1 it competes iff $v_1 < w$ with $r < w$, so types in $[r, w)$ are deterred only in the signalling equilibrium.

Ref: Amaro de Matos §8.2.1, pp. 206–207; Fishman (1988).

10. Jegadeesh–Chowdhry: no pooling equilibrium — Amaro de Matos 8.1.5 #8, p. 205

What is asked. With toehold $\alpha_m = 0.3$, types $Z_L = 40$ (40%) and $Z_H = 80$ (60%), and success probability $P(\pi) = \ln \pi$, show that the pooling equilibrium (both types hold α_m and bid Z_M) breaks down.

Approach. A pooling equilibrium fails if some type can deviate to an action that the other type would not mimic and that the deviating type strictly prefers. Here the low type deviates to a lower bid Z_L together with a smaller toehold α', chosen so that the high type is exactly indifferent (the cheapest credible signal). This uses results 1 and 2 of Jegadeesh–Chowdhry: the cost of a lower toehold is the same for both types, but a lower bid helps the low type more.

Notation.

- $G(\pi, \alpha; Z) = P(\pi)[0.5Z - (0.5 - \alpha)\pi]$: bidder's value (8.6), bid premium π, toehold α, type Z.
- $P(\pi) = \ln \pi$: increasing and concave (Assumption 8.1.4) and positive for $\pi > 1$.
- $Z_M = \mathbb{E}[Z]$: pooled bid (free-rider condition binding: $\pi = \mathbb{E}[Z \mid \alpha, \pi]$).

Step 1 — Pooled bid. With no information in α or π, the free-rider condition requires $\pi = \mathbb{E}[Z]$:

$$Z_M = (0.4)(40) + (0.6)(80) = 16 + 48 = 64. \tag{23}$$

Step 2 — Values in the pool. With $\ln 64 = 4.15888$,

$$\begin{aligned} G(64, 0.3; 80) &= \ln(64)\,[0.5(80) - (0.5 - 0.3)(64)] \\ &= 4.15888\,[40 - 12.8] = 4.15888(27.2) = 113.1216, \end{aligned} \tag{24}$$

$$\begin{aligned} G(64, 0.3; 40) &= \ln(64)\,[0.5(40) - (0.2)(64)] \\ &= 4.15888\,[20 - 12.8] = 4.15888(7.2) = 29.944. \end{aligned} \tag{25}$$

The low type pays $64 > 40$ for a gain worth 40 to it: it is subsidizing the high type.

Step 3 — The deviation that the high type will not mimic. The low type bids $\pi = Z_L = 40$ (which shareholders accept if they believe the bidder is the low type) with toehold α' such that the

high type is indifferent:

$$\begin{aligned} G(40,\alpha';80) &= G(64,0.3;80) \\ \ln(40)\,[40-(0.5-\alpha')40] &= 113.1216 \\ 3.688879\,(20+40\alpha') &= 113.1216 \\ 20+40\alpha' &= \frac{113.1216}{3.688879} = 30.6656 \\ \alpha' &= \frac{30.6656-20}{40} = 0.26664. \end{aligned} \tag{26}$$

Any α slightly below α' makes mimicking strictly unattractive for the high type.

Step 4 — The low type gains. Its value from the deviation is

$$\begin{aligned} G(40,\alpha';40) &= \ln(40)\,[0.5(40)-(0.5-0.26664)(40)] \\ &= 3.688879\,[20-9.3344] = 3.688879(10.6656) = 39.344 > 29.944. \end{aligned} \tag{27}$$

Answer (#8). The low type strictly prefers to bid 40 with a toehold $\alpha' = 0.2666 < 0.3$ (value 39.34 versus 29.94 in the pool), and the high type does not imitate it (value 113.12 either way); so the pooling equilibrium does not survive. The toehold works as a signal: a *smaller* toehold credibly signals a lower valuation.

Step 5 — Why the signal separates. Result 2: $\partial G/\partial\alpha = \pi P(\pi)$ does not depend on Z, so reducing the toehold from 0.3 to α' at bid 40 costs both types the same, $40\ln(40)(0.3-0.26664) = 4.92$. Result 1: $\partial[G(64)-G(40)]/\partial Z = 0.5[P(64)-P(40)] > 0$, so bidding low is worth more to the low type. Equal costs and unequal benefits sort the types.

Intuition. In Jegadeesh–Chowdhry the bid premium *rises* with the toehold: a bidder who buys many shares before the offer reveals a high valuation, and must then bid high. A low-valuation bidder deliberately buys fewer shares than it could, to be believed when it bids low.

Common mistakes.

- Choosing α' to make the *low* type indifferent; the binding constraint is that the high type does not mimic.
- Using Z_M as the low type's bid after deviating; the point of the deviation is the lower bid.

Book answer. Matches the book's answer (pp. 274–275): $Z_M = 64$, $G = 113.1216$ and 29.944, $\alpha' = 0.2667$; the book prints $G(40, 0.2667; 40) = 39.337$, against 39.344 here with α' unrounded (rounding of α' and of the logarithms).

Ref: Amaro de Matos §8.1.4, pp. 201–204; Jegadeesh and Chowdhry (1994).

11. Takeover defenses: a flip-over pill whose new shares vote — Tirole 11.1, p. 441

What is asked. Redo Tirole's poison-pill analysis (§11.5.2.4) when the shares issued to non-tendering shareholders after a hostile takeover carry votes: what fraction the raider must buy, the shareholders' indifference condition, the raider's premium and profit, and the optimal pill.

Approach. Free-riding logic (Grossman–Hart, Tirole §11.5.1): with a continuum of shareholders the success probability β makes them indifferent between tendering and holding. The pill is a transfer from the raider to the non-tendering holders after success. Voting new shares add one element: they dilute the raider's *votes*, so the raider needs a larger fraction of the old shares to keep control after the pill is triggered. Tirole's footnote 26 says the analysis is "basically unchanged"; we verify that and quantify the change.

Notation.

- v, $\hat{v}$: investor value per old share under incumbent and raider, $\hat{v} - v \equiv 1$; $\hat{w}$: raider's private benefit.
- m: new shares received free by each untendered old share after a successful takeover (Tirole's example: $m = 1$).
- κ': fraction of old shares the raider must buy; Δ_u: gain per untendered old share; Δ_r: dilution per raider share.
- P: premium over v; β: probability of success.

Step 1 — Control threshold. After success the raider holds κ' old shares and the others hold $(1-\kappa')(1+m)$ shares, all voting. He keeps a majority iff

$$\begin{aligned} \frac{\kappa'}{\kappa' + (1+m)(1-\kappa')} &\geq \frac{1}{2} \\ 2\kappa' &\geq \kappa' + (1+m)(1-\kappa') \\ \kappa'(2+m) &\geq 1+m \\ \kappa' &\geq \frac{1+m}{2+m}. \end{aligned} \tag{28}$$

With non-voting new shares the threshold is $\kappa = \frac{1}{2}$; with $m = 1$ it becomes $\kappa' = 2/3$.

Step 2 — Cash-flow split at the threshold. Total cash-flow shares are $\kappa' + (1+m)(1-\kappa')$, which at $\kappa' = (1+m)/(2+m)$ equals $2(1+m)/(2+m)$. A raider share is worth $\hat{v}(2+m)/[2(1+m)]$ and an untendered old share (with its m new ones) $(1+m)\hat{v}(2+m)/[2(1+m)] = \hat{v}(2+m)/2$:

$$\Delta_u = \frac{(2+m)\hat{v}}{2} - \hat{v} = \frac{m\hat{v}}{2}, \qquad \Delta_r = \hat{v} - \frac{(2+m)\hat{v}}{2(1+m)} = \frac{m\hat{v}}{2(1+m)}. \tag{29}$$

The pill is a pure transfer: $\kappa'\Delta_r = \frac{1+m}{2+m} \cdot \frac{m\hat{v}}{2(1+m)} = \frac{m\hat{v}}{2(2+m)} = (1-\kappa')\Delta_u$, so

$$\kappa'(\Delta_r + \Delta_u) = (1-\kappa')\Delta_u + \kappa'\Delta_u = \Delta_u. \tag{30}$$

Step 3 — Tendering indifference. A holder compares $v + P$ with keeping a share worth $\hat{v} + \Delta_u$ if the raid succeeds and v otherwise:

$$\begin{aligned} \beta(\hat{v} + \Delta_u) + (1-\beta)v &= v + P \\ \beta(\hat{v} - v + \Delta_u) &= P \\ \beta &= \frac{P}{1+\Delta_u}. \end{aligned} \tag{31}$$

Step 4 — Raider's profit. He buys κ' shares at $v + P$, which are worth $\hat{v} - \Delta_r$ to him on success and v on failure:

$$\begin{aligned} \pi &= \beta\hat{w} + \kappa'[\beta(\hat{v} - \Delta_r) + (1-\beta)v - (v+P)] \\ &= \beta\hat{w} + \kappa'[\beta(1-\Delta_r) - P] \\ &= \beta\hat{w} + \kappa'[\beta(1-\Delta_r) - \beta(1+\Delta_u)] \quad \text{(using (31))} \\ &= \beta\hat{w} - \beta\kappa'(\Delta_r + \Delta_u) \\ &= \beta(\hat{w} - \Delta_u) \quad \text{(using (30))}. \end{aligned} \tag{32}$$

This is Tirole's formula with Δ replaced by Δ_u. If $\hat{w} > \Delta_u$ the raider sets $\beta = 1$, i.e. $P = 1 + \Delta_u$, and earns $\hat{w} - \Delta_u$; otherwise he does not bid.

Step 5 — Numbers. $v = 2$, $\hat{v} = 3$, $\hat{w} = 2$, $m = 1$. Voting: $\kappa' = 2/3$, $\Delta_u = 1(3)/2 = 1.5$, $\Delta_r = 3/4 = 0.75$, check $\frac{2}{3}(0.75) = \frac{1}{3}(1.5) = 0.5$; premium $P = 1 + 1.5 = 2.5$ and profit $2 - 1.5 = 0.5$. Non-voting (Tirole's footnote 27): $\kappa = \frac{1}{2}$, $\Delta = \hat{v}\,m/(2+m) = 3/3 = 1$, $P = 2$, profit $2 - 1 = 1$.

Step 6 — Optimal pill. With $\hat{w}$ known, the target sets $\Delta_u^* = \hat{w}$, extracting the whole private benefit ($P = 1 + \hat{w}$, $\pi = 0$), exactly as with non-voting shares. Only the instrument changes: voting new shares deliver $\Delta_u = m\hat{v}/2$ per untendered share versus $m\hat{v}/(2+m)$, so fewer new shares are needed, $m = 2\hat{w}/\hat{v} = 4/3$ (with $\kappa' = 0.7$) instead of $m = 2\hat{w}/(\hat{v} - \hat{w}) = 4$ in the example.

Answer (11.1). With voting new shares the raider must buy $\kappa' = (1+m)/(2+m) > \frac{1}{2}$ of the shares; the equilibrium is otherwise unchanged: $\beta = P/(1+\Delta_u)$, $\pi = \beta(\hat{w} - \Delta_u)$, bid $P = 1 + \Delta_u$ when $\hat{w} > \Delta_u$, optimal pill $\Delta_u^* = \hat{w}$. A given number of new shares is a stronger defense (larger Δ_u), and the pill also raises the number of shares that must be tendered.

Intuition. Whatever the pill does to votes, it is a transfer from the raider to the holdouts on success. Free riding prices that transfer into the premium, so the raider pays it on every share he buys; net of the transfer, he keeps only his private benefit minus the holdouts' gain. Voting rights change how many shares he must buy, not that accounting.

Common mistakes.

- Keeping $\kappa = \frac{1}{2}$: with voting new shares, a bare majority of old shares loses control once the pill is triggered.
- Writing the raider's loss per share as Δ_u (the holdouts' gain); only at $\kappa = \frac{1}{2}$ are the two equal.

Book answer. No answer in the book; Tirole's footnote 26 ("basically unchanged") is consistent with the result.

Ref: Tirole §11.5.1–11.5.2, pp. 431–434.

12. Hansen: every cash offer is dominated by a stock offer — Amaro de Matos 8.2.6 #4, p. 223

What is asked. For any cash offer C that the acquirer would make (non-negative expected gain), construct a stock offer that is accepted by exactly the same target types and leaves the acquirer with a higher expected wealth.

Approach. The book's construction: $\beta^* = C/[x + w(C)]$. Two steps: (i) show that under Hansen's assumption $dw/dv < (x+w)/v$ the target types who accept β^* are exactly those who accept C, namely $v \le C$; (ii) subtract the two expected wealths and sign the difference. The economics: stock pays the target in a currency whose value moves with $w(v)$, which shifts part of the lemons cost back to the target.

Notation.

- $v \in [v_l, v_h]$, distribution F, density f; $w(v) \ge v$, increasing; x: acquirer's value (known).
- C: feasible cash offer, $E(\Pi_r \mid C) = F(C)[E(w \mid C) - C] \ge 0$, with $E(w \mid C) = \mathbb{E}[w(v) \mid v \le C]$.
- $\beta^* = C/[x + w(C)]$; W^{cash}, W^{stock}: acquirer's expected wealth.

Step 1 — The acceptance rule is the same. Type v accepts β^* iff $\beta^*[x + w(v)] \ge v$, i.e. iff $h(v) \equiv v/[x + w(v)] \le \beta^* = h(C)$. Differentiating h (quotient rule),

$$h'(v) = \frac{[x + w(v)] - v\,w'(v)}{[x + w(v)]^2} > 0 \iff w'(v) < \frac{x + w(v)}{v}, \tag{33}$$

which is Hansen's assumption. So h is strictly increasing and $h(v) \le h(C) \iff v \le C$: the stock offer is accepted with probability $F(C)$, by the same types as the cash offer.

Step 2 — The two expected wealths. From (8.14) and the stock formula,

$$W^{\text{cash}} = F(C)[E(w \mid C) - C] + x, \tag{34}$$

$$W^{\text{stock}} = F(C)(1 - \beta^*)[E(w \mid C) + x] + [1 - F(C)]x. \tag{35}$$

Step 3 — The difference. Expand (35) and subtract (34):

$$\begin{aligned} W^{\text{stock}} - W^{\text{cash}} &= F(C)\{(1-\beta^*)[E(w \mid C) + x] - x\} + x - F(C)[E(w \mid C) - C] - x \\ &= F(C)\{E(w \mid C) + x - \beta^*[E(w \mid C) + x] - x - E(w \mid C) + C\} \\ &= F(C)\{C - \beta^*[E(w \mid C) + x]\}. \end{aligned}$$

Substitute $C = \beta^*[x + w(C)]$ (definition of β^*):

$$\begin{aligned} W^{\text{stock}} - W^{\text{cash}} &= F(C)\{\beta^*[x + w(C)] - \beta^*[E(w \mid C) + x]\} \\ &= F(C)\,\beta^*\,[w(C) - E(w \mid C)]. \end{aligned} \tag{36}$$

Step 4 — Sign. $E(w \mid C)$ averages w over $v \le C$, and w is increasing, so $E(w \mid C) \le w(C)$ with strict inequality when w is strictly increasing and F has mass below C. Since $F(C) > 0$ and $\beta^* > 0$ for an offer that is ever accepted, (36) is positive. The stock offer also satisfies the acquirer's participation constraint, because its gain exceeds the cash gain, which is ≥ 0.

Step 5 — Illustration. $v \sim U[0, 100]$, $w(v) = 1.2v + 30$, $x = 200$ (Hansen's condition holds: $1.2 < (230 + 1.2v)/v$). For $C = 60$: $F = 0.6$, $E(w \mid C) = 1.2(30) + 30 = 66$,

$$\text{cash gain} = 0.6(66 - 60) = 3.6, \qquad \beta^* = \frac{60}{200 + 102} = 0.19868,$$

$$\text{stock gain} = 3.6 + 0.6(0.19868)(102 - 66) = 3.6 + 4.29 = 7.89.$$

The optimal cash offer solves $w(C) - C - F(C)/f(C) = 1.2C + 30 - C - C = 0$, so $C^* = 37.5$ with gain $0.375(52.5 - 37.5) = 5.625$; the matching stock offer does better again by $0.375\,\beta^*(75 - 52.5)$.

Answer (#4). For any feasible C, the stock offer $\beta^* = C/[x + w(C)]$ is accepted by the same types ($v \le C$) and raises the acquirer's expected wealth by $F(C)\beta^*[w(C) - E(w \mid C)] \ge 0$. Cash offers are (weakly) dominated; strictly so when w is strictly increasing.

Intuition. With cash the marginal accepting type $v = C$ is paid C, and so is every lower type. With stock, the payment $\beta^*[x + w(v)]$ is lower for lower types, because the shares they receive are worth less when the asset they bring is worth less. The acquirer saves $\beta^*[w(C) - w(v)]$ on each inframarginal type, which is (36) on average.

Common mistakes.

- Matching the stock offer by value at the mean type instead of at the marginal type $v = C$; then the acceptance sets differ.
- Forgetting Hansen's monotonicity assumption: without it $v/[x + w(v)]$ may be non-monotone and the acceptance set of a stock offer is not an interval.

Book answer. The book only refers to page 80 of Hansen (1987) (p. 277); the proof above completes the "simple algebraic exercise" stated on p. 210.

Ref: Amaro de Matos §8.2.2, pp. 208–210; Hansen (1987).

13. Fishman (1989): face value needed to signal, cash versus debt — Amaro de Matos 8.2.6 #8, p. 223

What is asked. Show that, to signal that v_1 is at least a given level, bidder 1 needs a smaller face value if it pays in cash than if it pays with debt: $p_c(v_1) < p_d(v_1)$.

Approach. A signal works if lower types are unwilling to send it; the face value needed to signal $v_1 \ge r$ is the most that type r is willing to offer, $p_c(r)$ with cash and $p_d(r)$ with debt (above it type r prefers not to preempt). Compare the two closed forms. The economics: cash is a *costlier* signal per unit of face value, because it is paid even in the bad state, when the target is worth $v_1^- < p$ to the bidder; debt is refused in the bad state and costs nothing there.

Notation.

- $\gamma \in (0,1)$: probability of the good signal s^+; $v_1^{\pm}$, $\tilde{v}_2^{\pm}$: valuations in each state, $v_i^+ > v_i^-$.
- $p_d(v_1) = \mathbb{E}\min[\max(v, \tilde{v}_2^+), v_1^+]$; $p_c(v_1) = \gamma\, p_d(v_1) + (1-\gamma)v_1^-$.
- Model facts: $v \le p \le v_1^+$ (so $v_1^+ \ge v$) and $\min\{v_1^-, p\} < v$ with $p \ge v$ (so $v_1^- < v$).

Step 1 — Write the difference. Subtract p_d from p_c:

$$\begin{aligned} p_c(v_1) - p_d(v_1) &= \gamma p_d(v_1) + (1-\gamma)v_1^- - p_d(v_1) \\ &= (1-\gamma)v_1^- - (1-\gamma)p_d(v_1) \\ &= (1-\gamma)\,[v_1^- - p_d(v_1)]. \end{aligned} \tag{37}$$

Step 2 — Bound p_d from below. For every realization, $\max(v, \tilde{v}_2^+) \ge v$ and $v_1^+ \ge v$, so $\min[\max(v, \tilde{v}_2^+), v_1^+] \ge v$. Taking expectations, $p_d(v_1) \ge v$.

Step 3 — Bound v_1^- from above. In the bad state the debt offer (face $p \ge v$) is worth $\min\{v_1^-, p\} < v$. If $v_1^- \ge v$ then $\min\{v_1^-, p\} \ge v$, a contradiction; so $v_1^- < v$.

Step 4 — Sign. Combining, $v_1^- < v \le p_d(v_1)$, so $v_1^- - p_d(v_1) < 0$; with $1-\gamma > 0$, (37) is negative:

$$p_c(v_1) < p_d(v_1). \tag{38}$$

For $\gamma = 1$ (the target's signal is uninformative about the bad state), $p_c = p_d$.

Step 5 — Illustration. $\gamma = 0.6$, $v = 0.6$, $\tilde{v}_2^+ \sim U[0,1]$, $v_1^+ = 0.9$, $v_1^- = 0.4$:

$$\begin{aligned} p_d &= (0.6)(0.6) + \frac{0.9^2 - 0.6^2}{2} + (0.9)(0.1) = 0.36 + 0.225 + 0.09 = 0.675, \\ p_c &= (0.6)(0.675) + (0.4)(0.4) = 0.405 + 0.16 = 0.565, \end{aligned}$$

and $p_c - p_d = (0.4)(0.4 - 0.675) = -0.11$.

Answer (#8). $p_c(v_1) - p_d(v_1) = (1-\gamma)[v_1^- - p_d(v_1)] < 0$ because $p_d \ge v > v_1^-$: signalling a given valuation needs a lower face value with cash than with debt.

Intuition. A signal is credible when it is expensive for those who should not send it. Paying cash commits the bidder to overpay in the bad state (pay p for an asset worth v_1^-), a cost that a low type fears more; so a smaller cash offer is as informative as a larger debt offer. This is why, in Fishman's model, preemptive (high-valuation) bidders pay cash, the opposite of Hansen's link between cash and an undervalued acquirer.

Common mistakes.

- Comparing the *values* of the offers to the target instead of their face values.
- Forgetting that debt is worthless to the bidder's cost in the bad state because the target rejects it.

Book answer. Matches the book's answer (pp. 277–278): the inequality reduces to $(\gamma-1)[p_d - v_1^-] < 0$, equality when $\gamma = 1$.

Ref: Amaro de Matos §8.2.3, pp. 212–214; Fishman (1989).

14. Shleifer–Vishny with an arbitrary resale price — Tirole 14.2, p. 515

What is asked. Redo §14.2.2 when the distressed firm's assets sell at any per-unit price $P \in [0, \rho_0]$ rather than $P = \rho_0$: the entrepreneurs' utilities (the analogues of α and κ), the borrowing capacity, the interaction between the two firms' investments and the symmetric equilibrium.

Approach. Tirole's continuous-investment moral-hazard model (pledgeable income ρ_0, NPV ρ_1 per unit) applied to an industry: assets of a distressed firm are worth something only to the other firm, whose investors can pay at most the pledgeable income ρ_0 (the *redeployability* and *fire-sale* logic). A lower P moves surplus from the seller's to the buyer's investors; competitive lending passes each lender's surplus to its entrepreneur.

Notation.

- I_i, A_i: firm i's investment and net worth; x: probability a firm is productive; μ, ν: Table 14.1 conditional probabilities, $x(1-\mu) = (1-x)(1-\nu)$ (14.3).
- $\rho_1 = p_H R$: per-unit NPV; $\rho_0 = p_H(R - B/\Delta p) < 1$: per-unit pledgeable income; $\rho_1 - \rho_0 = p_H B/\Delta p$: entrepreneur's per-unit rent.
- $P \in [0, \rho_0]$: per-unit resale price; $\hat{\alpha}$, $\hat{\kappa}$: coefficients of $U_{bi} = \hat{\alpha} I_i + \hat{\kappa} I_j$.

Step 1 — Who gets what in a transaction. When firm j is distressed and i productive (probability $x(1-\mu)$), lender i pays PI_j and raises entrepreneur i's stake so that $(\Delta p)\delta R_{bi} = BI_j$; the acquired assets yield expected income $\rho_1 I_j$, of which $(\rho_1 - \rho_0)I_j$ is the entrepreneur's rent and $\rho_0 I_j$ goes to lender i. Per unit:

$$\text{seller's investors: } P, \qquad \text{buyer's investors: } \rho_0 - P, \qquad \text{buyer's entrepreneur: } \rho_1 - \rho_0. \tag{39}$$

Step 2 — Entrepreneur's utility. Lenders break even, so entrepreneur i gets the NPV of everything her firm does: own operations, the sale of her assets when she alone is distressed (probability $(1-x)(1-\nu)$), and the purchase of j's assets:

$$\begin{aligned} U_{bi} &= [x\rho_1 - 1]I_i + (1-x)(1-\nu)PI_i + x(1-\mu)[(\rho_1 - \rho_0) + (\rho_0 - P)]I_j \\ &= [x\rho_1 + (1-x)(1-\nu)P - 1]I_i + x(1-\mu)(\rho_1 - P)I_j \equiv \hat{\alpha} I_i + \hat{\kappa} I_j. \end{aligned} \tag{40}$$

Comparing with $\alpha = x\rho_1 + (1-x)(1-\nu)\rho_0 - 1$ and $\kappa = x(1-\mu)(\rho_1 - \rho_0)$,

$$\hat{\alpha} = \alpha - (1-x)(1-\nu)(\rho_0 - P), \qquad \hat{\kappa} = \kappa + x(1-\mu)(\rho_0 - P), \tag{41}$$

and by (14.3) the two corrections are equal, so $\hat{\alpha} + \hat{\kappa} = \alpha + \kappa$: bargaining power only redistributes surplus between the firms for given investments.

Step 3 — Borrowing capacity. Lender i's breakeven now includes its surplus $\rho_0 - P$ on acquisitions:

$$x\rho_0 I_i + (1-x)(1-\nu)PI_i + x(1-\mu)(\rho_0 - P)I_j = I_i - A_i. \tag{42}$$

Collecting the I_i terms on the right,

$$\begin{aligned} A_i + x(1-\mu)(\rho_0 - P)I_j &= I_i[1 - x\rho_0 - (1-x)(1-\nu)P] \\ I_i &= \frac{A_i + x(1-\mu)(\rho_0 - P)I_j}{1 - x\rho_0 - (1-x)(1-\nu)P} \\ &= \frac{A_i + x(1-\mu)(\rho_0 - P)I_j}{1 + (1-x)(1-\nu)(\rho_0 - P) - \rho_0[x + (1-x)(1-\nu)]}. \end{aligned} \tag{43}$$

For $P < \rho_0$, $\partial I_i/\partial I_j = x(1-\mu)(\rho_0 - P)/[1 - x\rho_0 - (1-x)(1-\nu)P] > 0$: investments are *strategic complements* (a cheap future acquisition raises what lender i will lend today), while firm i's own collateral is worth less (P instead of ρ_0). For $P = \rho_0$ this is (14.7).

Step 4 — Symmetric equilibrium. With $A_1 = A_2 = A$ and $I_1 = I_2 = I$, (43) gives

$$I[1 - x\rho_0 - (1-x)(1-\nu)P - x(1-\mu)(\rho_0 - P)] = A.$$

Using $x(1-\mu) = (1-x)(1-\nu)$, the P terms cancel: $-(1-x)(1-\nu)P-(1-x)(1-\nu)\rho_0+(1-x)(1-\nu)P$, so

$$I = \frac{A}{1-\rho_0[x+(1-x)(1-\nu)]} = kA, \tag{44}$$

independent of P: the lower collateral value and the cheaper acquisition offset each other.

Step 5 — Investment decision. Each entrepreneur invests if $\hat{\alpha} > 0$ (her own investment's marginal value; $\hat{\kappa} I_j$ is a windfall she does not control). Because $\hat{\alpha} \leq \alpha$, the threshold correlation $\hat{\nu}^*$ where $\hat{\alpha} = 0$ is below ν^*: lower seller bargaining power shrinks the region where firms invest, although total surplus per symmetric firm $(\hat{\alpha}+\hat{\kappa})I$ is unchanged when they do.

Step 6 — Illustration. $x = 0.8$, $\nu = 0.3$ ($\mu = 1 - 0.14/0.8 = 0.825$), $\rho_1 = 1.5$, $\rho_0 = 0.6$, $P = 0.3$, $A = 1$: $\alpha = 1.2 + 0.084 - 1 = 0.284$, $\kappa = 0.8(0.175)(0.9) = 0.126$, $\hat{\alpha} = 0.284 - 0.14(0.3) = 0.242$, $\hat{\kappa} = 0.126 + 0.042 = 0.168$, and $\hat{\alpha} + \hat{\kappa} = 0.41 = \alpha + \kappa$. The denominator of (43) is $1 - 0.48 - 0.042 = 0.478$, the slope $0.042/0.478 = 0.0879$, and $I = 1/(1 - 0.6 \times 0.94) = 1/0.436 = 2.294$ for any P, NPV $0.41 \times 2.294 = 0.940$. With $x = 0.6$ instead, $\nu^* = 1 - 0.1/(0.4 \times 0.6) = 0.583$ but $\hat{\nu}^* = 1 - 0.1/(0.4 \times 0.3) = 0.167$.

Answer (14.2). $U_{bi} = \hat{\alpha} I_i + \hat{\kappa} I_j$ with $\hat{\alpha} = \alpha - (1-x)(1-\nu)(\rho_0 - P)$, $\hat{\kappa} = \kappa + x(1-\mu)(\rho_0 - P)$, $\hat{\alpha} + \hat{\kappa} = \alpha + \kappa$; $I_i = [A_i + x(1-\mu)(\rho_0 - P)I_j]/[1 - x\rho_0 - (1-x)(1-\nu)P]$, increasing in I_j when $P < \rho_0$; in symmetric equilibrium $I = A/\{1 - \rho_0[x + (1-x)(1-\nu)]\}$, independent of P.

Intuition. The resale price is a transfer between the two industry members. It changes each firm's incentive to invest (selling cheap hurts, buying cheap helps) and makes each firm's debt capacity depend on the other's size; but in a symmetric industry every firm is equally likely to be the seller and the buyer, so the transfers wash out of debt capacity and total value.

Common mistakes.

- Leaving lender i's acquisition surplus $(\rho_0 - P)I_j$ out of the breakeven condition; it is what makes capacity depend on I_j.
- Using $\alpha + \kappa$ to decide investment; an individual firm invests on $\hat{\alpha}$, the windfall term does not depend on its own choice.

Book answer. Matches the book's answer (p. 618): $\hat{\alpha}$, $\hat{\kappa}$, the capacity formula and the P-independence of symmetric investment.

Ref: Tirole §14.2.2, pp. 500–503.

15. Liquidity management and acquisitions — Tirole 14.3, p. 515

What is asked. (i) The safe firm's optimal credit line and acquisition cutoff when the retooling cost ρ is random and privately observed, given the price P: show $P + \rho^* = 1$. (ii) The distressed seller's pricing problem, the equilibrium $P < 1$, and how P responds to a larger credit line. (iii) The limit when F collapses to a point $\bar{\rho}$.

Approach. Chapter 5's liquidity–scale trade-off: a unit of liquidity reserved for a future need costs investment scale today (multiplier $1/(1-\rho_0)$), and a credit line caps which shocks are met. Here the "shock" is an acquisition opportunity of cost $(P+\rho)J$. The seller is a monopolist facing an unknown buyer's maximum willingness to pay, so it trades off price against the probability of a sale (monotone hazard rate gives a unique price).

Notation.

- I, J: safe and risky firms' investments; A: safe firm's net worth; $1 - x$: probability the risky firm is distressed.
- $\rho \sim F$ on $[0, \infty)$, density f, f/F decreasing; $G(\rho) \equiv F(\rho)/f(\rho)$, so $G' > 0$.

- L: credit line per unit of J; $\rho_0 J$: cash from a seasoned offering; the acquisition is feasible iff $(P+\rho)J \le (\rho_0 + L)J$, so $\rho^* = \rho_0 + L - P$.

Step 1 — (i) NPV and breakeven. The acquisition, when it happens, yields NPV $\rho_1 - (P+\rho)$ per unit and pledgeable income $\rho_0 - (P+\rho)$ per unit to investors (who fund it through the line and the offering). Following Chapter 5,

$$U_b^s = (\rho_1 - 1)I + (1-x)J \int_0^{\rho^*} [\rho_1 - (P+\rho)]\, dF(\rho), \tag{45}$$

$$\rho_0 I + (1-x)J \int_0^{\rho^*} [\rho_0 - (P+\rho)]\, dF(\rho) = I - A. \tag{46}$$

Solving (46) for I (move $\rho_0 I$ to the right and divide by $1-\rho_0 > 0$):

$$I = \frac{A - (1-x)J \int_0^{\rho^*} [(P+\rho) - \rho_0]\, dF(\rho)}{1-\rho_0}. \tag{47}$$

Substituting (47) in (45):

$$U_b^s = (\rho_1 - 1)\frac{A - (1-x)J \int_0^{\rho^*} [(P+\rho) - \rho_0]\, dF}{1-\rho_0} + (1-x)J \int_0^{\rho^*} [\rho_1 - (P+\rho)]\, dF. \tag{48}$$

Step 2 — (i) Optimal cutoff. Differentiate (48) in ρ^* (Leibniz rule: each integral contributes its integrand at ρ^* times $f(\rho^*)$), and write $s = P + \rho^*$:

$$\frac{\partial U_b^s}{\partial \rho^*} = (1-x)Jf(\rho^*)\Big[- \frac{(\rho_1 - 1)(s - \rho_0)}{1-\rho_0} + \rho_1 - s\Big].$$

Setting the bracket to zero and multiplying by $1-\rho_0$:

$$\begin{aligned}
(\rho_1 - s)(1-\rho_0) &= (\rho_1 - 1)(s - \rho_0) \\
\rho_1 - \rho_1\rho_0 - s + s\rho_0 &= \rho_1 s - \rho_1\rho_0 - s + \rho_0 \\
\rho_1 + s\rho_0 &= \rho_1 s + \rho_0 \\
\rho_1 - \rho_0 &= s(\rho_1 - \rho_0) \\
s &= 1.
\end{aligned}$$

The bracket is positive for $s < 1$ and negative for $s > 1$ (it is linear in s with slope $-(\rho_1 - 1)/(1-\rho_0) - 1 < 0$), so this is a maximum:

$$P + \rho^* = 1, \qquad \text{credit line } L^* = \rho^* + P - \rho_0 = 1 - \rho_0 \text{ per unit of } J. \tag{49}$$

Answer (i). The safe firm secures a credit line L^*J with $\rho_0 + L^* = P + \rho^* = 1$: it acquires iff the total unit cost of buying, $P+\rho$, is at most 1, the unit cost of growing internally ("make or buy"). Numerically, with F exponential ($\lambda = 2$), $\rho_0 = 0.6$, $\rho_1 = 1.5$, $x = 0.8$, $J = A = 1$ and $P = 0.4$, a grid search over ρ^* maximizes U_b^s at $\rho^* = 0.6 = 1 - P$.

Step 3 — (ii) Seller's problem. The seller does not observe ρ and anticipates the line L; a sale occurs iff $\rho \le \rho_0 + L - P$. Its assets have no other use, so it maximizes expected revenue per unit:

$$\max_P \; P\, F(\rho_0 + L - P). \tag{50}$$

The first-order condition (with $\rho^* = \rho_0 + L - P$) is

$$F(\rho^*) - Pf(\rho^*) = 0 \iff P = \frac{F(\rho^*)}{f(\rho^*)} = G(\rho^*). \tag{51}$$

In equilibrium $L = L^* = 1 - \rho_0$, so $\rho^* = 1 - P$ and P solves $\phi(P) \equiv -P + G(1-P) = 0$. At $P = 0$, $\phi = G(1) > 0$; at $P = 1$, $\phi = -1 + G(0) = -1 < 0$ (as $F(0) = 0$); and $\phi'(P) = -1 - G'(1-P) < 0$ because $G' > 0$ by the monotone hazard rate. Hence a unique $P \in (0,1)$.

Step 4 — (ii) Larger credit line. Differentiate $-P + G(\rho_0 + L - P) = 0$ in L:

$$-\frac{dP}{dL} + G'\Big(1 - \frac{dP}{dL}\Big) = 0 \;\Rightarrow\; \frac{dP}{dL} = \frac{G'}{1+G'} \in (0,1). \tag{52}$$

The cutoff rises, $d\rho^*/dL = 1 - dP/dL = 1/(1+G') > 0$: the seller captures part of a larger line but the probability of a sale still increases. Example $F(\rho) = 1 - e^{-2\rho}$: $G(\rho) = (e^{2\rho} - 1)/2$, $G' = e^{2\rho}$; the equilibrium is $P = 0.6040$, $\rho^* = 0.3960$, $F(\rho^*) = 0.547$, and $dP/dL = e^{0.792}/(1 + e^{0.792}) = 0.688$ (confirmed by finite differences).

Answer (ii). The risky firm maximizes $PF(\rho_0 + L - P)$; the equilibrium price solves $P = F(1-P)/f(1-P)$, unique and in $(0,1)$; a higher anticipated L raises P with pass-through $G'/(1+G') < 1$, so the probability of a deal rises too.

Step 5 — (iii) Spike at $\bar\rho$. Informally (as in the book): a price whose cutoff $1 - P$ stays below $\bar\rho$ sells with probability near 0, so the seller does better by setting a cutoff just above $\bar\rho$; a cutoff that stays above $\bar\rho$ gives $Pf \approx 0$ and $F \approx 1$, violating the first-order condition $F = Pf$ unless the cutoff converges to $\bar\rho$. So $1 - P \to \bar\rho$ and $F(\rho^*) \to 1$. Explicit sequence: F uniform on $[\bar\rho - \varepsilon, \bar\rho + \varepsilon]$ ($f/F = 1/(\rho - \bar\rho + \varepsilon)$ decreasing). Revenue $P \cdot \min\{(1 - P - \bar\rho + \varepsilon)/(2\varepsilon), 1\}$ increases in the cutoff up to $\rho^* = \bar\rho + \varepsilon$ when $1 - \bar\rho > 3\varepsilon$, so $P = 1 - \bar\rho - \varepsilon$ and $F(\rho^*) = 1$; as $\varepsilon \to 0$, $P \to 1 - \bar\rho$. Example $\bar\rho = 0.5$, $\varepsilon = 0.05$: $P = 0.45$, sale with probability 1.

Answer (iii). $P + \bar\rho = 1$ in the limit and $F(\rho^*) \to 1$: the price equals the buyer's make-or-buy indifference price, the competitive outcome, and a deal is (almost) sure.

Intuition. The buyer hoards exactly the liquidity that makes buying as cheap as building; the seller, knowing this, prices like a monopolist against the buyer's uncertain retooling cost. With little uncertainty the monopolist extracts everything up to the make-or-buy price and the deal always happens; with more uncertainty it prices some deals away, so trade is inefficiently low.

Common mistakes.

- Forgetting the opportunity cost of liquidity: hoarding reduces I through (47); ignoring it gives the wrong cutoff $\rho^* = \rho_1 - P$.
- Treating L as observed by the seller out of equilibrium; the seller prices on the *anticipated* line.

Book answer. Matches the book's answer (pp. 618–619): $\rho^* = 1 - P$, $\rho_0 + L^* = 1$; P maximizes $F(\rho_0 + L^* - P)P$, unique in $(0,1)$; $0 < dP/dL < 1$; informal limit argument for (iii).

Ref: Tirole §14.2.5.3, pp. 506–508; §5.3.1, pp. 207–209.